



ANDERSON JUNIOR COLLEGE
2018 JC 2 PRELIMINARY EXAMINATION

NAME: _____

PDG: ____/17

CHEMISTRY

8873/02

Paper 2 Structured Questions

12 September 2018

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use							
Paper 1 (33%)	Paper 2 (67%)						
	Section A				Section B		Total
	Q1	Q2	Q3	Q4	Q5*	Q6*	
/30							/ 80
				Final marks		/ 100	
				Grade			

* Circle the question you have attempted

This document consists of **21** printed pages.

Section A

Answer **all** the questions in this section, in the spaces provided.

- 1 Ethanol, $\text{CH}_3\text{CH}_2\text{OH}$, is a renewable biofuel which is made from the starch in corn grain or from the glucose in sugar cane.

(a) What is meant by the term *standard enthalpy change of combustion*?

.....
[2]

- (b) When 1 g of ethanol was burned under a container of water, it was found that 100 g of water was heated from 20°C to 65°C . The process was known to be only 65 % efficient.

Use these data and values from the *Data Booklet* to calculate the enthalpy change of combustion of ethanol.

[3]

- (c) Fig 1.1 shows an energy cycle involving ethanol.

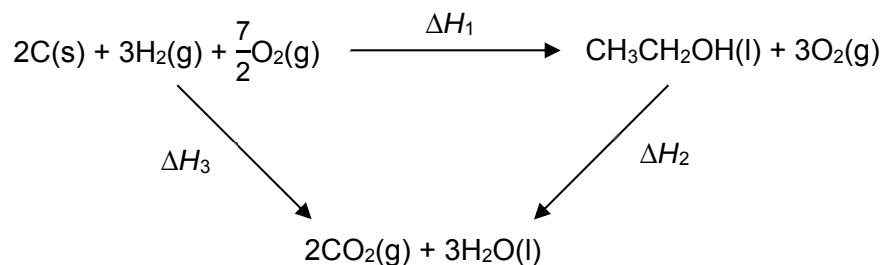


Fig 1.1

- (i) Name the enthalpy change represented by ΔH_1 .

.....[1]

- (ii) Given that the enthalpy change of combustion of carbon and hydrogen are $-393.5 \text{ kJ mol}^{-1}$ and $-285.8 \text{ kJ mol}^{-1}$ respectively, use the energy cycle given above and your answer to (b) to calculate ΔH_1 .

[3]

- (d) In some countries, ethanol is blended with petrol for use in cars. Ethanol blended petrol can significantly reduce the greenhouse gas emissions. The carbon dioxide released by a vehicle when ethanol is burned is offset by the carbon dioxide captured when the feedstock crops are grown to produce ethanol.

Petrol which contains 10% ethanol by volume is named E10 and it is the most widely used around the world. On average, 2.33 kg of carbon dioxide is emitted for every one litre of petrol combusted in a car engine. In contrast, every one litre of ethanol combusted in the car engine reduces mass of carbon dioxide emitted by 99 % compared to petrol.

According to oil industry data, cars in Singapore consumed about 6,453,600 barrels of petrol in 2017. Each barrel is approximately 159 litres.

- (i) Calculate the total mass of CO_2 , in kg, emitted by cars in Singapore in 2017.

[1]

- (ii) Calculate the mass of CO_2 , in kg, emitted for every litre of ethanol combusted in the car engine.

[1]

- (iii) Calculate the mass of CO_2 , in kg, emitted for every litre of E10 petrol combusted in the car engine.

[1]

- (iv) Hence, calculate the reduction in mass of CO_2 , in kg, emitted if E10 was used by cars in Singapore in 2017 instead of regular petrol.

[1]

- (e) Poly(vinyl alcohol) (PVA) is used to make film laundry bags which are used in hospitals to minimise the contact hospital workers have with contaminated clothing and bedding. The dirty items are put into these special bags, which are then placed directly into the washing machine. As the bags are water-soluble, they dissolve and are washed down the drain with the dirty water.

Fig 1.2 shows part of the PVA structure with three repeat units.

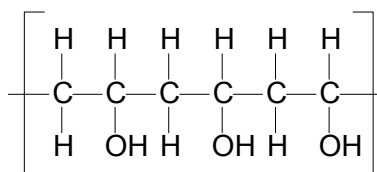


Fig 1.2

- (i) Draw the full displayed formula of the monomer of PVA.

[1]

- (ii) State the reagent and conditions required to convert the monomer of PVA to ethanol.

..... [1]

- (iii) State the type of intermolecular forces that can be formed between PVA polymer and water molecules.

..... [1]

- (iv) Copy the structure of PVA in the space below and use it to illustrate the type of attraction between PVA polymer and **one** water molecule.

[2]

- (v) When borax, $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ was gradually added to PVA solution and stirred, the mixture thickens and eventually forms slime. When dissolved in solution, borax forms the borate ion, $\text{B}(\text{OH})_4^-$, which has the structure given in Fig 1.4.

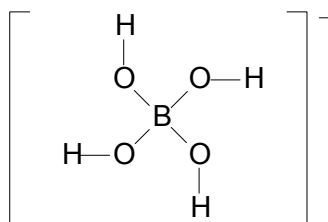


Fig 1.4

Suggest a reason why the presence of borate ions causes mixture to thicken and form slime.

.....
 [1]

- (vi) PVA cannot be synthesised directly from its monomer. Instead, it is formed from poly(vinyl acetate). Part of the structure of poly(vinyl acetate) is shown in Fig 1.3.

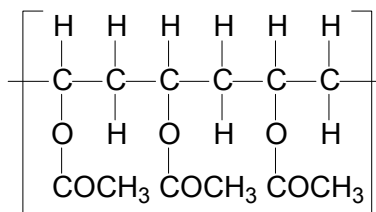


Fig 1.3

Suggest the reagent and condition required to form PVA from poly(vinyl acetate).

..... [1]

[Total: 20]

- 2 (a) Poly(ethene) is a *thermoplastic*.

Explain what you understand of the word in *italics*.

.....
.....[1]

- (b) Ethene can be polymerised in slightly different ways to make low-density poly(ethene) (LDPE) and high-density poly(ethene) (HDPE). These polymers have different physical properties because of the structure of their molecules. LDPE is flexible whereas HDPE is relatively more rigid.

- (i) Describe the structure and bonding in LDPE and HDPE and explain how they affect their physical properties.

.....
.....
.....
.....
.....
.....
.....
.....[4]

- (ii) Suggest a use for each polymer in consideration of its physical properties.

HDPE:

LDPE:[2]

- (c) An experiment was carried out to determine the relative oxidising power of three unknown halogens X_2 , Y_2 and Z_2 . Table 2.1 shows the results of experiments in which the halogens X_2 , Y_2 and Z_2 were added to separate aqueous solutions containing X^- , Y^- and Z^- ions.

Table 2.1

	X^- (aq)	Y^- (aq)	Z^- (aq)
X_2		no reaction	no reaction
Y_2	brown solution formed		orange solution formed
Z_2	brown solution formed	no reaction	

- (i) Using the results given in Table 2.1, suggest the identity of X_2 , Y_2 and Z_2 .

X_2 :

Y_2 :

Z_2 :

[2]

- (ii) Hence, state the most powerful oxidising agent among the halogens.

..... [1]

- (iii) Explain, in terms of ease of gain of electrons, why the element you have identified in (c)(ii) is the most powerful oxidising agent.

.....

 [2]

- (d) Three Group 17 hydrides were separately heated in an experiment. Their bond energy values and the observations of the experiment are given in Table 2.2.

Table 2.2

H-X	bond energy / kJ mol^{-1}	observation
H-Cl	431	no visible change
H-Br	366	brown fumes of Br_2
H-I	299	dense purple fumes of I_2 on gentle heating

- (i) Using the observations of the experiment, state the thermal stability of the Group 17 hydrides in decreasing order.

..... [1]

- (ii) Explain, in terms of their bond energies, the trend in the thermal stability of Group 17 hydrides.

.....

.....

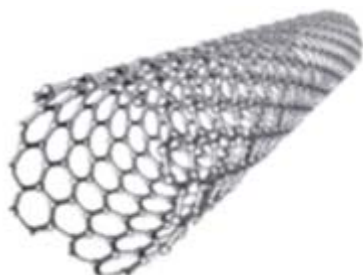
.....

.....

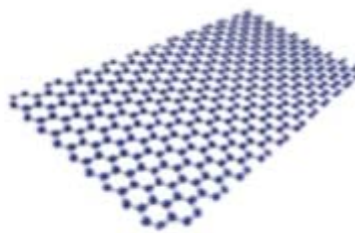
.....

.....[2]

- (e) A carbon nanotube (CNT) resembles a sheet of graphene rolled up to form a tiny cylindrical tube, with a diameter in the nanometer scale. They are used to manufacture professional tennis rackets and hockey sticks to make them lighter and more durable under constant forceful impacts by the players.



Carbon nanotube (CNT)



Graphene

- (i) Suggest one similarity and one difference in terms of physical property between CNT and graphene.

Similarity:

.....

Difference:

.....[2]

- (ii) State what is meant by the term “*nanometer scale*”.

.....[1]

- (iii) As nanotechnology continues to expand into every industrial sector, workers will be at an increased risk of exposure to new nanomaterials. Consumers also face increased exposure to nanomaterials as they are found in hundreds of products, ranging from cosmetics, to clothing, to industrial and biomedical applications. A number of recent research studies with rodents have shown that CNTs may pose a respiratory hazard to human beings.

State two other ways in which nanomaterials may enter the human body, other than through inhalation.

.....

.....[2]

[Total: 20]

- 3 (a) Beams of particles travelling at the same speed from different sources are subjected to an electric field. Fig 3.1 shows the experimental set-up where protons are found to be deflected through an angle of $+x^\circ$.

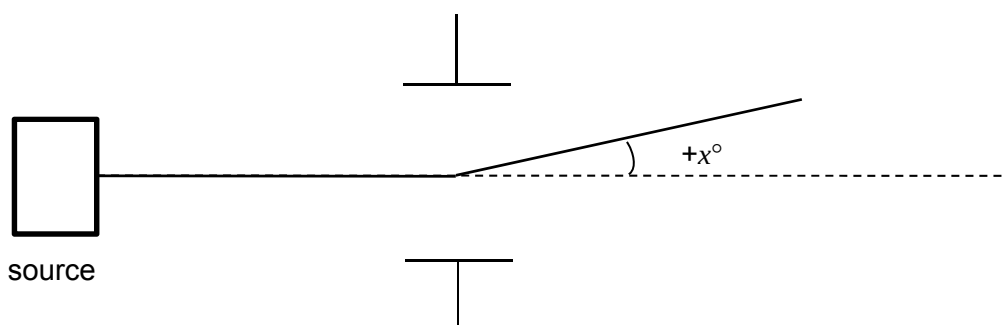


Fig 3.1

- (i) Indicate, in Fig 3.1, the polarity of the plates and explain your answer.

.....
 [2]

- (ii) Assuming an identical set of experimental conditions, state and explain the **angle** and **direction** of deflection for the following particles.

I: $^{16}\text{O}^{2-}$ ion

.....

II: ^4He nuclei

.....

III: ^2H atom

.....

 [5]

- (b) Fig 3.2 shows a scheme of the 17 lowest energy subshells, which can be used to show the order in which the subshells are filled by electrons (the Aufbau principle).

1s	2s	3s	4s	5s	6s	7s
	2p	3p	4p	5p	6p	
		3d	4d	5d		
			4f	5f		

Fig 3.2

- (i) List the order of filling subshells for an atom of a d–block element in the **fourth** period of the Periodic Table.
[1]
- (ii) Hence, write the electronic configuration of the copper atom.
[1]
- (iii) An electron of the copper atom undergoes excitation from the ground state. What is the lowest energy, empty orbital that the electron can be promoted to?
[1]

[Total: 10]

- 4 A laboratory technician found a bottle in the school laboratory that contained a white crystalline solid. The label on the bottle was smudged such that he could only make out the words “acid” and “104 g mol⁻¹”. To investigate, he decided to conduct a titration using 0.100 mol dm⁻³ sodium hydroxide.

He dissolved 2.29 g of the unknown acid in 250 cm³ of deionised water and pipetted 25.0 cm³ of this solution into a conical flask. He gradually added 0.100 mol dm⁻³ NaOH(aq) solution from a burette and monitored the pH of the reaction mixture in the conical flask using a pH meter after each addition.

Fig 4.1 shows the pH–volume added curve he obtained.

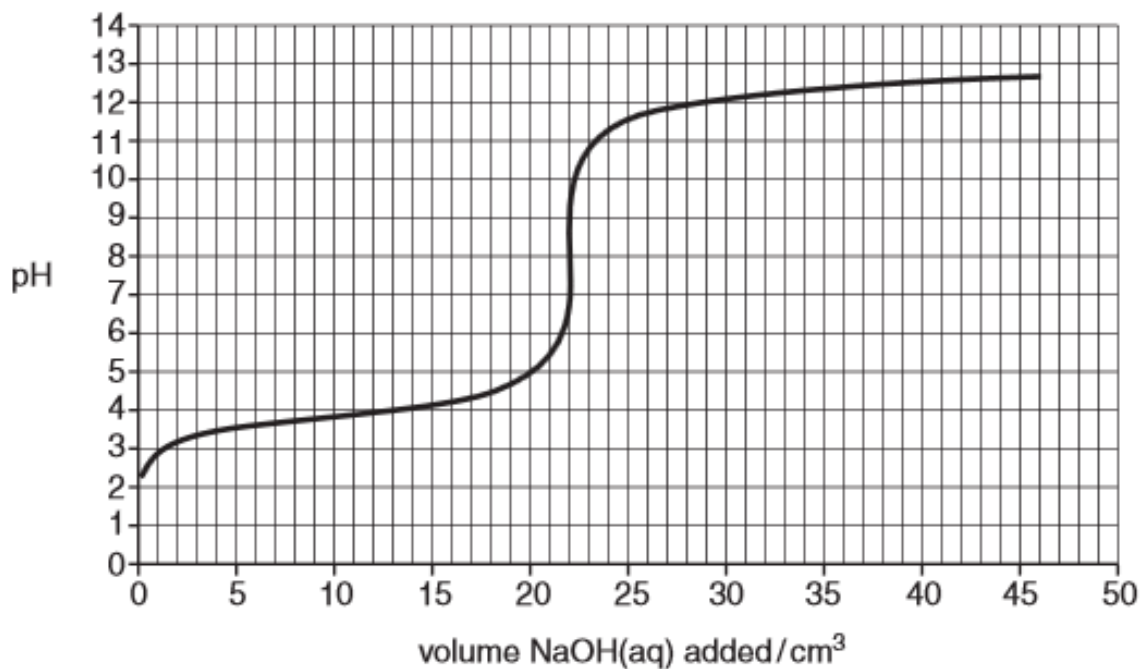


Fig 4.1

- (a) Calculate the number of moles of the unknown acid transferred to the conical flask.

[2]

- (b) Calculate the number of moles of sodium hydroxide that reacted with the unknown acid in the conical flask.

[1]

- (c) Hence, using your answers to (a) and (b), determine if the unknown acid is monobasic or dibasic.

[1]

- (d) Given that the initial pH is 2.2, explain why the unknown acid is a weak acid.

.....
.....[2]

- (e) Hence, or otherwise, deduce the acid dissociation constant, K_a of the unknown acid.

[2]

- (f) Show that the concentration of excess unreacted hydroxide ions when 45 cm³ of sodium hydroxide was added to the conical flask is 0.0329 mol dm⁻³.

[1]

- (g) Hence, calculate the pH of the solution after 45 cm³ of sodium hydroxide was added.

[1]

[Total: 10]

Section B

Answer **one** question from this section, in the spaces provided.

- 5 (a) The oxides of Period 3 elements Na_2O , Al_2O_3 and P_4O_{10} show different physical and chemical properties. Some information on these oxides is given in Table 5.1.

Table 5.1

oxide	melting point / °C	pH of the resulting solution from the reaction with H_2O
A	2040	7
B	24	2
C	1132	13

- (i) Describe the reactions, if any, of the oxides Na_2O and P_4O_{10} with water. Include the approximate pH value of any resulting solutions, and write equations for any reactions that occur.

.....

.....

.....

.....

.....

.....

.....[4]

- (ii) Hence, suggest the identity of **A**, **B** and **C**.

A	B	C

[2]

- (iii) Explain, in terms of structure and bonding, why **A** has a higher melting point than **C**.

.....

.....

.....

.....

.....

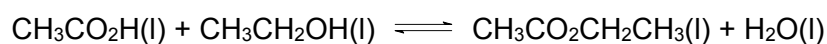
.....[3]

- (b) SO_2 is another oxide of a Period 3 element.

Draw a dot-and-cross diagram to show the bonding in a molecule of SO_2 .

[1]

- (c) When ethanoic acid reacts with ethanol, in the presence of concentrated sulfuric acid, to form ethyl ethanoate and water, the following equilibrium is established.



- (i) Explain the purposes of adding concentrated sulfuric acid.

.....

[2]

- (ii) Write an expression for the equilibrium constant, K_c , for the reaction between ethanoic acid and ethanol.

[1]

- (iii) Equimolar amounts of ethanoic acid and ethanol were mixed and at equilibrium, 1.00 mol of ethyl ethanoate is present.

Given that the value of K_c is 4.0, determine the initial amount (in mole) of ethanoic acid.

[2]

- (iv) Predict what would happen to the amount of ethyl ethanoate if sodium hydroxide was added to the reaction mixture at equilibrium. Explain your answer.

Prediction:

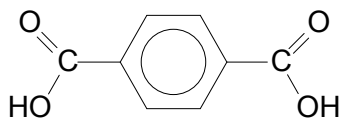
.....

Explanation:

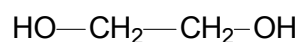
.....[2]

- (d) Polyesters are polymers formed from a dicarboxylic acid and a diol and they are mostly used in clothing, food packaging and plastic water and carbonated soft drinks bottles.

An example of a polyester is poly(ethylene terephthalate) (PET). Its monomer is synthesised from a condensation reaction between 1 mol of ethane-1,2-diol and 1 mol of benzene-1,4-dicarboxylic acid in the presence of a catalyst.



benzene-1,4-dicarboxylic acid



ethane-1,2-diol

- (i) Write a balanced equation for the formation of the PET monomer.

.....[1]

- (ii) Bottles that are made of PET cannot be used to store alkaline cleaning solution. Bottles that are made of poly(propene) are used instead.

Explain, in terms of the bonds present in both polymers, why the above statements are true.

.....

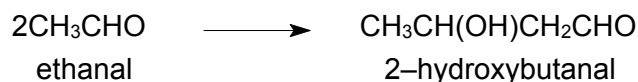
.....

.....

.....[2]

[Total: 20]

- 6 (a) The Russian composer Borodin was also a research chemist who discovered a reaction in which two ethanal molecules combine to form an aldol, 2-hydroxybutanal.



- (i) Suggest the type of reaction which takes place in the formation of 2-hydroxybutanal from ethanal. Explain your answer.

.....

[2]

- (ii) Draw the displayed formula of 2-hydroxybutanal. Indicate clearly, in your answer, one bond angle which is 105° and one which is 120° .

[3]

On heating, 2-hydroxybutanal, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CHO}$, forms compound **X**, $\text{C}_4\text{H}_6\text{O}$, which exists as a pair of isomers.

When compound **X** was heated with acidified $\text{K}_2\text{Cr}_2\text{O}_7$, compound **Y**, $\text{C}_4\text{H}_6\text{O}_2$ is formed. **Y** reacts with sodium carbonate to produce carbon dioxide gas.

- (iii) Suggest the type of reaction which occurs upon heating 2-hydroxybutanal.

.....[1]

- (iv) Draw the full structural formulae of the pair of isomers of **X**. Indicate clearly the type of isomerism by labelling the isomers.

[2]

(v) Suggest the structural formula of Y.

[1]

- (b) An experiment was carried out to determine the order of reaction with respect to bromoethane in the following hydrolysis reaction with $0.10 \text{ mol dm}^{-3} \text{ NaOH(aq)}$.

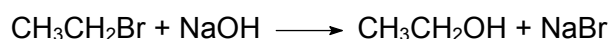


Table 6.1 shows the data that was obtained from this experiment.

Table 6.1

time / min	$[\text{CH}_3\text{CH}_2\text{Br}] / \text{mol dm}^{-3}$
0	0.0100
30	0.0084
60	0.0070
90	0.0058
120	0.0049
150	0.0041
180	0.0034

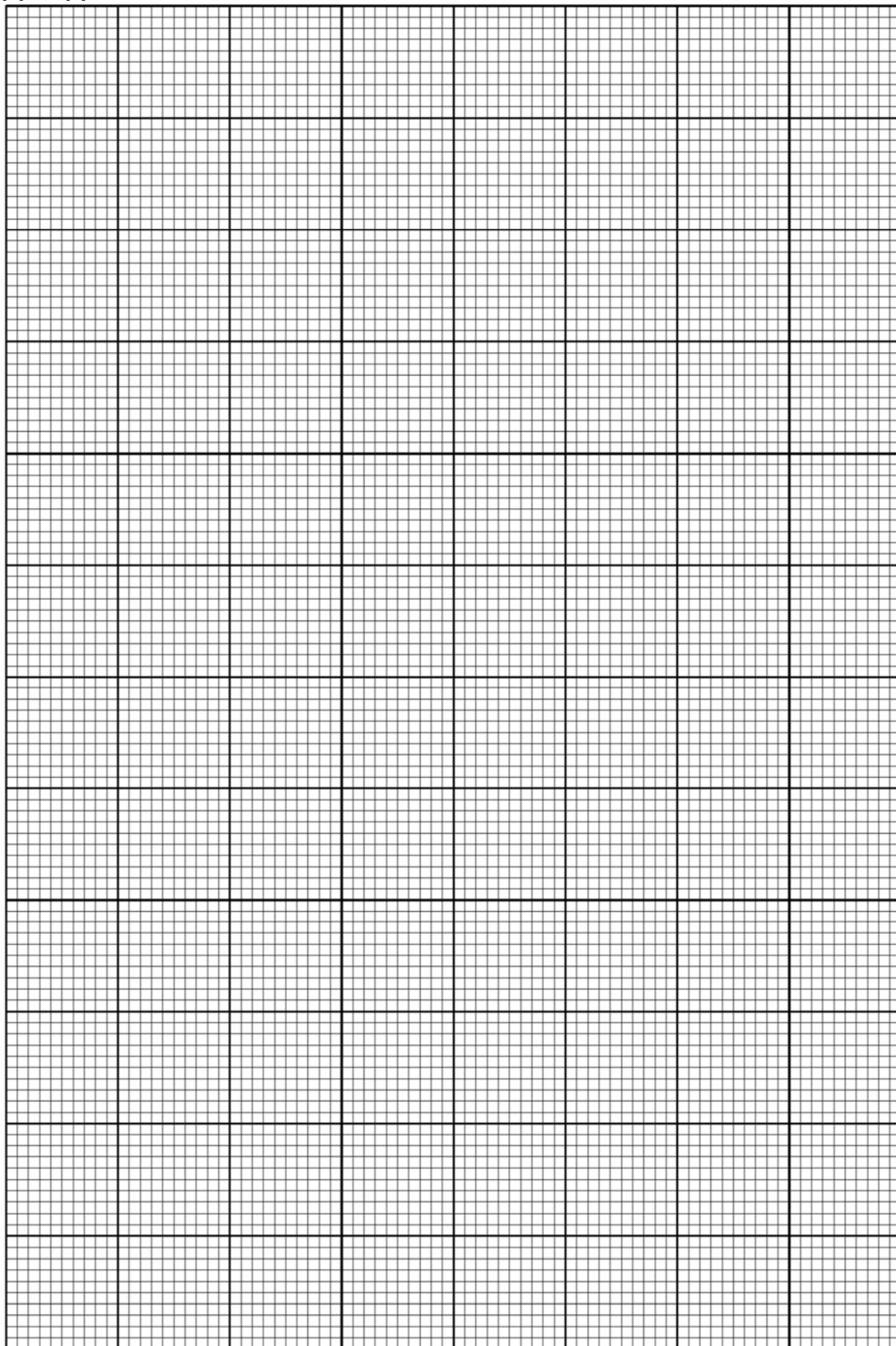
- (i) The concentration of NaOH(aq) used in the experiment was in large excess relative to that of bromoethane. Suggest a reason why this is so.

.....
[1]

- (ii) Plot a suitable graph using the grid provided on page 19 to determine the order of reaction with respect to the bromoethane. Show your working clearly.

order with respect to bromoethane:[3]

6 (b) (ii)



- (iii) The order of reaction with respect to sodium hydroxide is found to be first order. Hence, construct a rate equation for the hydrolysis reaction.

.....[1]

- (iv) Use your graph to determine the initial rate of reaction.

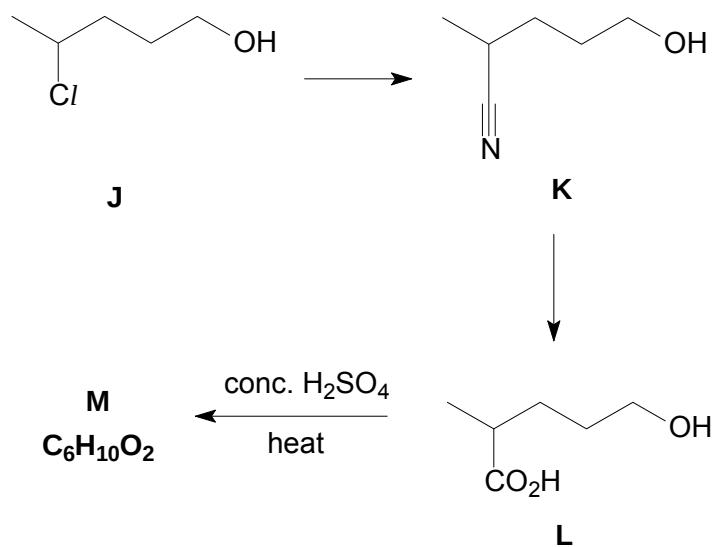
[1]

- (v) Using your answer in (b)(iv) and other relevant experiment data given, calculate a value for the rate constant, stating its units.

[1]

- (c) Use the table of characteristic values for infra-red absorption in the *Data Booklet* to answer this question.

Compound **J** undergoes a series of transformation as shown.



Infra-red absorptions can be used to identify functional groups in organic compounds. For example, compound **J** shows absorptions at 700 – 800 cm⁻¹, but not compounds **K** and **L**.

- (i) Identify **two** infra-red absorption ranges that will be shown by **J**, **K** and **L**.
[1]
- (ii) Identify an infra-red absorption range that will be shown by **L** but not by **J** and **K**.
[1]
- (iii) The infra-red spectrum of **M** shows absorption in two ranges, 1050–1330 cm⁻¹ and 1710–1750 cm⁻¹.

Suggest a structural formula for **M** and name the type of reaction which occurs in the last transformation step from **L** to **M**.

Type of reaction: [2]

[Total:20]